Damien MAÎTRE

MSc Engineering Student — CFD, HPC & Numerical Methods

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PROFILE

MSc engineering student specializing in industry, structural mechanics and high performance computing, with hands-on experience across CFD, data analysis and infrastructure engineering. I'm seeking an engineering internship in the energy industry, tackling engineering R&D, grid management or project finance, starting March 2027.

EDUCATION

September 2026 – February 2027
MSc Physics engineering
Student exchange at Politecnico di Milano, Italy
International economics, Energy engineering, High performance computing, Infrastructure finance, Neural Networks.

September 2023 – June 2027
MSc Engineering — Fluid dynamics, Structural mechanics, Prototyping
École nationale des ponts et chaussées (ENPC), Paris
Structural mechanics and large-structure design, FEM applied to structural analysis, Machine learning, Metallurgy, Complete fluid dynamics module, HPC, Data science.

- **Tools:** OpenFOAM, StarCCM+, Abaqus, OpenSeesPy, gmsh
- **Projects:** Underwater sea-glider simulation and prototype ; closed-wing aircraft design and optimization ; wooden suspension bridge design, verification and construction.

RESEARCH PROJECTS

Complete CFD developper-first suite
Personal project — OpenFOAM, Gmsh (forked), Python, C++
Open and user-friendly aerodynamical loads studies, tailored for very high $\mathcal{R}e$ axisymmetric problems.

Structural FEM & Eurocode Verification suite
Personal project — OpenSeesPy, Python
End to end structure design, visualization and verification tools.

Data engineering projects
Python, PyTorch, Tensorflow, Numpy, C++ & handmade frameworks
Five years of history working on data analysis and machine learning projects within student teams. City-scale modelisation for mobility and urban optimization ; Genetic algorithm reproducing the swimming behavior of jellyfishes ; Data reporting for large scale engineering studies.

PROFESSIONAL EXPERIENCE

February 2026 – July 2026
Infrastructure engineering internship — TRACTEBEL FRANCE
CDG Express project, Saint-Ouen

- Construction compliance verification for the CDG Express, a 2.5B€ high-speed rail project,
- Structural analysis and Eurocode verifications of civil engineering structures,
- Review of structural calculation reports and as-built site surveys,
- Drafting technical reports and recommendations for the client.

July 2025 – January 2026
Data & Industry Project Manager — SNCF Voyageurs
Industry & digital solutions, Saint-Denis

- Management of a 150k€ digital project within the rolling stock maintenance industry,
- Coordination of 15 3D artists across the country's maintenance plants,
- Project ownership and reporting to internal clients (plant directors, Industrial Director, etc.),
- End-to-end procurement and tendering process.

TECHNICAL SKILLS

CFD & FEM	OpenFOAM, StarCCM+, Abaqus, OpenSeesPy, Gmsh
Programming	10+ years of experience with Python, C++, UNIX environments and various other programming languages.
Prototyping	Additive manufacturing, CAD, ARDIUNO

LANGUAGES

French — Native · **English** — Proficient (C1) · **Italian** — Working proficiency (B2) · **German** — Working proficiency (B2)

ACTIVITIES & COMMUNITY INVOLVEMENT

Rescuer — French Red Cross
Volunteer first-aid responder
First-aid responder at public events (marathons, festivals, Paris-Plage) and on-call with Paris Fire Brigade.

Director — Student Musical Ensemble
BridgeWay Comédie Musicale, Champs-Sur-Marne
Directed an ensemble of 75 artists and a creative/technical leadership team of 12 with 4.5k€ budget.

General Secretary — Student Association
Union des Elèves de l'Ecole Nationale des Ponts et Chaussées
Management of a team of 15 for professional and social events with 1,000+ attendees, Administrative management of a registered French non-profit (Loi 1901).